WT

1) Describe the importance of semantic elements in web development.

a) Semantic elements are HTML elements that clearly describe their meaning to both the browser and the developer.
Examples: include <header>, <nav>, <section>, <article>, <footer> etc.

1. They improve the readability of the code.
2. Help search engines understand webpage content.
3. Increase accessibility for Screen readers and disabled users.
4. Increase accessibility for screen readers and disabled users
5. provide better structure to web pages.
6. Make maintainence & debugging easier
7. Improve website ranking in search engines.

2. Explain the steps to implement tailwind css.

→ Tailwind css is a utility - first css framework used to design modern responsive websites

Steps to implement tailwind CSS :-

1. Install Node.js on the system
2. create a new project folder
3. Initialize npm using npm init -y.
4. Install Tailwind css using npm install -D tailwind css
5. Generate configuration file using npx tailwind css. init.

6. create an input css file and include Tailwind directives (@tailwind base; @tailwind components;).
7. Link the generated css file to the HTML file.
8. Use Tailwind utility classes in HTML

Ex: class = "bg- blue- 500 text-white (p-u")".

③ Describe how data persistance works in local storage and session storage.

A) Local storage and session storage are part of the web storage API used to store the data in the browser.

1. localstorage stores data with no expiration time
2. Data remains even after closing the browser.
3. Session storage stores data only for one session.
4. Data is deleted when the browser tab is closed
5. Data is stored in key - value pairs.
6. Methods used: SetItem() ,getItem(), removeItem(), clear().
7. storage capacity is about 5-10MB
8. mainly used to store user preferences, login. info and temporary data

Unit - 2

④ Illustrate prototypal inheritance in Javascript with an example.

→ prototypal inheritance allows objects to inherit properties and methods from another object.

→ In Javascript, every object has a prototype.

Example :

Javascript

```
function Person (name){
  this.name = name ;
}
person.protype.greet = function (){
console.log ("Hello "+ this.name);
};
const student = new person ("Harika");
student.greet();
```

⑤ Discuss Array Object methods with examples.

- push() - Adds element to the end.

  Ex :- let arr = [1,2,3]

  arr.push(4);

  console.log(arr); // [1,2,3,4].

- pop() - Removes the last element from the array.

  Ex :- arr.pop();

  console.log (arr); //[1,2,3]

- shift() - Removes a new array by applying a function to each element

- filter() -
- foreach()
- reduce().

⑥. Briefly explain the difference between let and const when declaring variables in Javascript.

Feature	Let	const.
Reassignment	. can be reassigned	. cannot be.
Redeclaration	. cannot be redeclared in same scope	. cannot be redeclared in same scope

Scope	• Block scoped	• Block scoped
Initialization	• can be declared without initializing	• cannot be redeclared in same scope.
Value change	• value can change later	• value cannot change
Use Case	• used when value may vary	• used for fixed values
Objects/ Arrays	• can modify and reassign.	• can modify properties but cannot reassign reference

Unit - 3

7) Differentiate the frontend and backend components of the MERN Stack.

Frontend	Backend.
• Built using React	• Built using Node.js & Express.
• Handles user interface	• Handles server logic
• Runs in browser	• Runs on server
• Manages UI Components	• Manages APIs
• Uses JSX	• Uses Javascript
• Communicates via HTTP	• Connects to MongoDB database
• focuses on design and interaction	• focuses on data processing
• visible to users	• Not directly visible
• client side processing.	• server side processing.

8) write a program to read and write a file using the fs module.

→ In Node.js, the fs module is used for file operations.

program :-

```
const fs = require ('fs');
//write a file.
fs.writeFileSync ('sample.txt', 'Hello world');
//Read from file
const data = fs.readFileSync ('sample.txt', 'utf8');
console.log (data);
```

(Q) what is REPL in Node.js? Explain.

→ Node.js provides REPL environment

→ REPL stands for:

- R - Read
- E - Evaluate
- P - print
- L - Loop

→ REPL is an interactive shell.

→ It reads user input.

→ REPL evaluates the javascript code.

→ prints the result.

→ Loops back for next input.

→ REPL is used for testing and debugging the code

→ started by typing node in terminal

→ Helps developers experiment quickly.